One Carrier, Two Readouts: Born, Lorentz, Bloch, and Killing

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One-sheet companion to *On Information Physics: From State Distinguishability to Causal Accumulation*

Origin. CAL is not a theory of quantum gravity. It is an information-physics engine for non-Markovian reality, built to lift self-attention to a field-theoretic form from first principles of cognition, causality and intelligence. Its derivations of known physics are unit tests: a law of causal memory that cannot recover established physics has no business modeling minds or societies.

Definition 1 (Information). Information := distinguishability of physical states.

Remark 1 (Difference and observation). To distinguish physical states is to observe a difference between them. Change is a difference between states along an ordered evolution; thus change is temporally ordered distinguishability. Observation here denotes physical distinguishability, not a required conscious observer.

Definition 2 (Causality). Causality := change in physical state, including its rate of change.

The Causal Accumulation Law (On Information Physics, Eq. 61)

$$T^{\mu\nu a}{}_{\rho\sigma}(x) = \underbrace{\alpha_{\mathrm{frac}}\, J^{\mu\nu a}{}_{\rho\sigma}(x)}_{\text{present source}} - (1-\alpha_{\mathrm{frac}}) \int_{J^-(x)} \underbrace{k(\tau; x, x')}_{\text{history kernel}} \underbrace{\Pi^{\mu\nu}{}_{\rho\sigma|\alpha\beta}{}^{\gamma\delta}(x, x')}_{\text{slot coupling}}$$
$$\times \underbrace{P^{\alpha}{}_{\alpha'}(x, x')\, P^{\beta}{}_{\beta'}(x, x')\, P_{\gamma}{}^{\gamma'}(x, x')\, P_{\delta}{}^{\delta'}(x, x')}_{\text{spacetime transport of the past to } x} \times \underbrace{\Phi^{a}{}_{b}(x)\, P^{b}{}_{c'}(x, x')}_{\text{internal channel action}} \quad J^{\alpha'\beta'c'}{}_{\gamma'\delta'}(x')\, d^4x'.$$

Read left to right: what the field is now (T) equals a share α_{frac} of the present source, minus a share $1-\alpha_{\mathrm{frac}}$ of the past. Only the causal past $J^-(x)$ counts. Each past contribution is weighted by how long ago it happened (k), carried to the present point so it can be compared there (the four P factors), and rotated in its internal channel (ΦP).

Definitions and symbols

Sym.	Meaning	Type	Eq.
$K(t,\tau)$, F	Memory kernel; aspect of past state	kernel; map	(5)
$M(t)$	History functional	path functional	(5)
τ_E, $\hbar_{\mathrm{eff}}$	Energy scale; action scale	positive reals	(9)
e_a	$e_0 = I_2$, $e_j = -i\sigma_j$	$M_2(\mathbb{C})$	(10)
J	Carrier $\sum_a Z_a e_a$	$\mathbb{C}\otimes\mathbb{H}\cong M_2(\mathbb{C})$	(11)
ϱ	Positive closure $JJ^\dagger$	Hermitian, $\succeq 0$	(12)
E_m, p_m	Positive effects; Born probabilities	$\succeq 0$; $[0,1]$	(13)
x^μ, $\eta_{\mu\nu}$	Closure four-vector; Minkowski metric	real	(16)
$\alpha \equiv \alpha_{\mathrm{frac}}$	Interaction depth; fractional order	real	(19)
I^α	Past-supported Weyl integral	operator	(19)
θ	Accumulated Wick angle $\pi\alpha/2$	angle	(21)
$\hat\varrho$, $\vec r$	Normalized closure; Bloch vector	state; $\lvert\vec r\rvert \le 1$	(29)
G	$SL(2,\mathbb{C})$ element	Lie group	(36)
κ	Killing form $\mathrm{tr}(\mathrm{ad}_A \mathrm{ad}_B)$	bilinear form	(49)
$T^{\mu\nu}$, w_i	Dyad sum; dyad weights	tensor; real	(57)
W, π^μ	Action weight; canonical momentum	positive fn.; field	(62)

1. History is more than its endpoints

The fundamental theorem of calculus recovers net change. A history functional can tell apart two trajectories with the same endpoints.

$$X(t) - X(t_0) = \int_{t_0}^{t} \dot X\, d\tau, \qquad M(t) = \int_{t_0}^{t} K(t,\tau)\, F\big(X(\tau)\big)\, d\tau.$$

Only two kernels make the present sufficient: the Dirac limit (forgetful) and the light-cone limit (faithful). Because K depends on the evaluation time, M is not additive under concatenation, so it is not a functor to any additive group; its carrier is the path category $\mathrm{Path}(S)$, outside the de Rham tiers.

2. Two weights in one channel

Each channel holds a Boltzmann weight and a Feynman weight. The substitution $1/k_BT \to 1/(i\hbar)$ turns one into the other, the move that makes the path integral a partition function with quantropy as its entropy.

$$Z_a = A_a e^{-H_a/\tau_E} + B_a e^{iS_a/\hbar_{\mathrm{eff}}}, \quad A_a + B_a = 1, \quad J = \textstyle\sum_a Z_a e_a \in M_2(\mathbb{C}).$$

3. The Born rule from positive closure

Closing the carrier with its own adjoint gives a positive matrix. Its trace adds up the channels; positive effects read probabilities off it.

$$\varrho = JJ^\dagger \succeq 0, \quad \tfrac{1}{2}\,\mathrm{tr}\,\varrho = \textstyle\sum_a |Z_a|^2, \quad p_m = \dfrac{\mathrm{tr}(E_m JJ^\dagger)}{\mathrm{tr}(JJ^\dagger)}.$$

For $J = \psi\eta^\dagger$ and $E_m = \phi_m\phi_m^\dagger$ this is $p_m = |\phi_m^\dagger\psi|^2/\psi^\dagger\psi$. The inputs are positive effects and trace evaluation; the squared amplitude is algebraic.

4. Lorentz from the determinant

The same matrix has a real Hermitian slice. Its determinant is the Minkowski interval, and $SL(2,\mathbb{C})$ congruence preserves it.

$$X = x^\mu\sigma_\mu, \quad \det X = (x^0)^2 - |\vec x|^2, \quad \det(GXG^\dagger) = \det X.$$

The closure sits in the future cone: $\det\varrho = |\det J|^2 \ge 0$, rank one is null, rank two is timelike. This is the $\mathbb{C}$ case of $\mathfrak{sl}(2,\mathbb{K}) \cong \mathfrak{so}(\dim\mathbb{K}+1, 1)$.

5. The Bloch ball is a slice of the cone

Normalize the trace to one and the future cone's cross-section is the Bloch ball.

$$\hat\varrho = \frac{\varrho}{\mathrm{tr}\,\varrho} = \tfrac{1}{2}\big(I_2 + \vec r\cdot\vec\sigma\big), \quad |\vec r| \le 1, \quad \det\hat\varrho = \tfrac{1}{4}\big(1 - |\vec r|^2\big).$$

Surface: pure, null closures. Interior: mixed, timelike closures. Rotations keep the normalization; boosts do not, so the Bloch section is a normalization, not a Lorentz-invariant slice. ***[Derived]*** *$2\sqrt{\det\varrho}/\,\mathrm{tr}\,\varrho = \sqrt{1-|\vec r|^2} = 1/\gamma = \sqrt{2S_L}$, the concurrence of a purification ($S_L = 1 - \mathrm{tr}\,\hat\varrho^2$); pure closures are null.*

6. The Killing form on the generator slice

On traceless anti-Hermitian generators the determinant is the Killing form of $\mathfrak{su}(2)$, up to scale.

$$\det A = |\vec y|^2 = -\tfrac{1}{2}\,\mathrm{tr}\,A^2 = -\tfrac{1}{8}\,\kappa(A,A), \qquad A = y^j e_j.$$

On all of $\mathfrak{u}(2)$ the Killing form misses the centre; the determinant restores it with opposite sign, $\det A = -(y^0)^2 + |\vec y|^2$. Trace gives probability, determinant gives the metric, Killing gives the internal symmetry: three readings of one carrier.

7. Real slices of one complex form

Over $\mathbb{C}$, $\det J = Z_0^2 + Z_1^2 + Z_2^2 + Z_3^2$ has no signature. Choosing which channels carry an i chooses the slice.

$\mathbb{H}$	$(+,+,+,+)$	quaternion norm
$i\mathfrak{u}(2)$ (Hermitian)	$(+,-,-,-)$	closures, Born, Minkowski
$\mathfrak{u}(2)$ (anti-Hermitian)	$(-,+,+,+)$	transport generators

Minkowski is the Euclidean form with the three vector channels rotated by i: the algebraic quarter turn of Wick rotation. ***[Derived]*** *Multiplication by i swaps the state and generator slices isometrically. The two pairings use two bars: Born uses †; the Lorentz pairing applies the bar $X \mapsto \mathrm{adj}\,X = \epsilon X^T \epsilon^T$ ($\epsilon = i\sigma_2$) once, $X\,\mathrm{adj}\,X = \det X\, I_2$; applying it twice returns X, giving back the Born pairing on the Hermitian slice.*

8. Order is kept, but lossily

Left-acting histories are remembered by $JJ^\dagger$ and erased by $J^\dagger J$.

$$J_{tx}J_{tx}^\dagger = U_t U_x \varrho U_x^\dagger U_t^\dagger \;\neq\; J_{xt}J_{xt}^\dagger, \qquad J_{tx}^\dagger J_{tx} = J_{xt}^\dagger J_{xt} = J^\dagger J.$$

The closure distinguishes some histories and identifies others: neither faithful nor forgetful. The commutator of two closures is anti-Hermitian, $[\varrho_1,\varrho_2]^\dagger = -[\varrho_1,\varrho_2]$ ***[Derived]****, so order-sensitive content lives on the generator slice; the Jarlskog CP invariant $\det[H_u, H_d]$ is of this form* ***[Proposed]****.*

9. Fractional accumulation keeps both readouts

The Weyl integral rescales the decay sector and rotates the phase sector. The rotation between them is the Wick angle.

$$I^\alpha e^{st} = s^{-\alpha}e^{st}, \quad I^\alpha Z_a = A_a\lambda_a^{-\alpha}e^{\lambda_a t} + B_a(i\omega_a)^{-\alpha}e^{i\omega_a t}, \quad \theta = \frac{\pi\alpha}{2}.$$

θ is a quarter turn at $\alpha = 1$ and a half turn at $\alpha = 2$. Since $\varrho^{(\alpha)} = (I^\alpha J)(I^\alpha J)^\dagger \succeq 0$, the Born and cone identities hold at every order.

10. Energy conditions from positivity

Build a rank-two tensor from closure vectors. Positive weights on future-causal vectors give the energy conditions with no extra input.

$$T^{\mu\nu} = \textstyle\sum_i w_i x_i^\mu x_i^\nu, \qquad T_{\mu\nu}u^\mu u^\nu = \sum_i w_i (x_i\cdot u)^2 \ge 0,$$
$$T^\mu{}_\nu u^\nu = \textstyle\sum_i w_i (x_i\cdot u)\, x_i^\mu, \qquad T^\mu{}_\mu = \sum_i w_i \det\varrho_i.$$

In the accumulation the weights are α_{frac} (present) and $(\alpha_{\mathrm{frac}}-1)k$ (history). With $k \ge 0$: $\alpha_{\mathrm{frac}} \ge 1$ gives weak, null, and dominant energy conditions automatically.

11. A weight gradient acts as friction [Derived]

Weight the action by a positive function and the Euler–Lagrange equations gain a term set by the log-gradient of the weight.

$$\partial_\mu\pi^\mu - \frac{\partial L}{\partial\Phi} = -(\partial_\mu \ln W)\,\pi^\mu.$$

$W = e^{\gamma t}$ gives the Caldirola–Kanai damped oscillator; $W = a^3$ gives Hubble friction. Exchanging the two integrals of the CAL action gives exactly this form, with $W(x) = \alpha_{\mathrm{frac}} + (\alpha_{\mathrm{frac}}-1)\int_{J^+(x)} k\, d^4z$ ***[Derived; finiteness of W open]****: uniform accumulation is invisible, a change in accumulation acts as friction, and matter exchanges energy with the weight, $\nabla^\mu(WT_{\mu\nu}) = L_m\nabla_\nu W$.*

12. Why closed systems forget [Proposed]

Finite capacity forces erasure at $k_BT\ln 2$ per bit, so a closed system reaches only the forgetful limit: physical memory is lossy.